The solid between $z = x^2 + y^2$ and $z = 5 - x^2 - y^2$

Problem

Use a graphing device to draw the solid enclosed by the paraboloids $z = x^2 + y^2$ and $z = 5 - x^2 - y^2$.

Notation used in this solution

- E is the solid, that is, the set of points (x, y, z) that lie between the two surfaces.
- r is the polar radius, $r^2 = x^2 + y^2$, and $R = \sqrt{5/2}$ is the radius of the circle where the two surfaces meet.
- $z = x^2 + y^2$ is the *lower bowl*. It opens upward, and its lowest point is the origin $(0, 0, 0)$.
- $z = 5 - x^2 - y^2$ is the *upper bowl*. It opens downward, and its highest point is $(0, 0, 5)$.
- $\mathrm{d}V$ is the volume element. In cylindrical coordinates $\mathrm{d}V = r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$, and in Cartesian coordinates $\mathrm{d}V = \mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$.

Part 1: the solid

1. Where the two surfaces meet

Both surfaces are bowls whose axis is the z-axis. The lower bowl rises as we move away from the axis, and the upper bowl falls. So they meet where the two heights are equal:

$$x^2 + y^2 = 5 - x^2 - y^2 \quad \Longleftrightarrow \quad 2(x^2 + y^2) = 5 \quad \Longleftrightarrow \quad x^2 + y^2 = \frac{5}{2}.$$

On that cylinder the height of either surface is

$$z = x^2 + y^2 = \frac{5}{2}.$$

So the two surfaces meet exactly on the circle

$$x^2 + y^2 = \frac{5}{2}, \qquad z = \frac{5}{2},$$

whose centre is $\left(0, 0, \frac{5}{2}\right)$ and whose radius is

$$R = \sqrt{\frac{5}{2}} = \frac{\sqrt{10}}{2} \approx 1.581139.$$

2. A description of the solid without a picture

The lower bowl is below the upper bowl exactly when

$$x^2 + y^2 \le 5 - x^2 - y^2 \quad \Longleftrightarrow \quad x^2 + y^2 \le \frac{5}{2}.$$

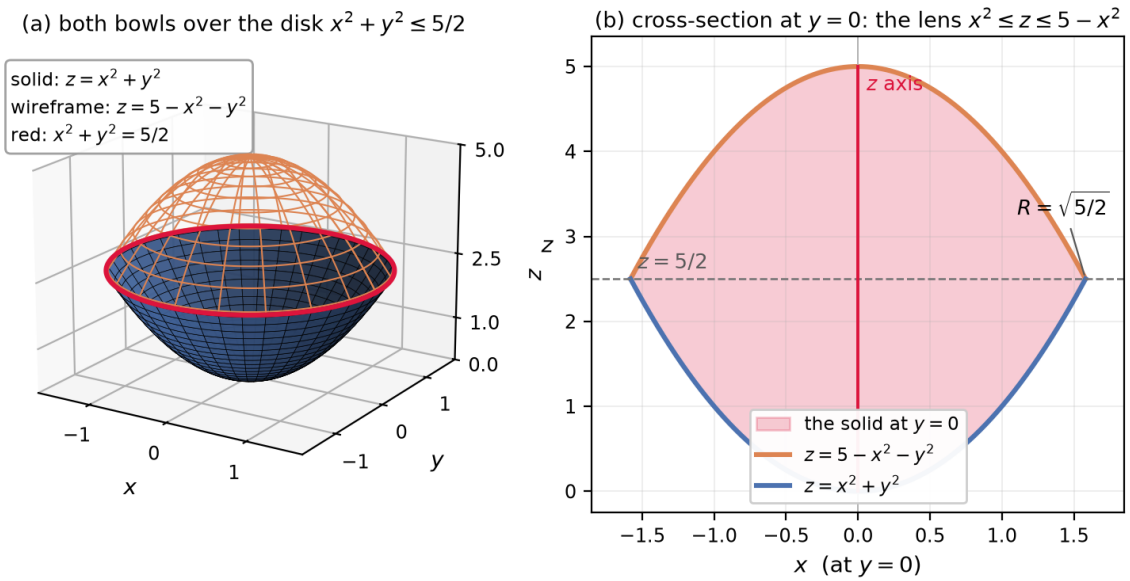


Figure 1: The solid enclosed by $z = x^2 + y^2$ and $z = 5 - x^2 - y^2$. (a) The lower bowl is drawn as a surface and the upper bowl as a wireframe, so both can be seen at once. The red circle is the rim $x^2 + y^2 = 5/2$ at height $z = 5/2$, where the two surfaces meet. (b) The cross-section at $y = 0$. The shaded lens is the part of the solid in that plane, and its widest width is $2R = 2\sqrt{5/2} \approx 3.162$.

Inside that disk the solid runs from the lower bowl up to the upper bowl. Therefore

$$E = \left\{(x, y, z) : x^2 + y^2 \le \frac{5}{2},\ \ x^2 + y^2 \le z \le 5 - x^2 - y^2\right\}.$$

Three sentences describe E completely:

1. E has the z-axis as an axis of symmetry, because both equations involve x and y only through $x^2 + y^2$.
2. At height z its horizontal cross-section is the disk $x^2 + y^2 \le r(z)^2$ with

$$r(z) = \sqrt{z}\ \ (0 \le z \le \tfrac{5}{2}), \qquad r(z) = \sqrt{5 - z}\ \ (\tfrac{5}{2} \le z \le 5).$$

   The cross-section is widest at $z = 5/2$, where $r = R$.
3. The boundary has two curved pieces, the lower bowl and the upper bowl, and one circle, the rim at $z = 5/2$. There is no flat face. In a horizontal slice the outline is a circle, and in a vertical slice through the axis the outline is two parabolas that meet at two points. The shape is a closed lens: two bowls glued along their common rim.

Figure 1 shows the two surfaces and the rim, together with the vertical cross-section at $y = 0$, where the solid is the region $x^2 \le z \le 5 - x^2$ between the two parabolas.

Quantity	Method	Value
$V(E)$	exact, $\pi\int_0^{5/2} z\,\mathrm{d}z + \pi\int_{5/2}^{5}(5-z)\,\mathrm{d}z$	19.634954084936
$V(E)$	exact, $2\pi\int_0^{R}(5-2r^2)r\,\mathrm{d}r$	19.634954084936
$V(E)$	Monte Carlo, box $\lvert x\rvert, \lvert y\rvert \le R$, $0 \le z \le 5$	19.642050
$V(E)$	Monte Carlo, cylinder $r \le R$, $0 \le z \le 5$	19.639392
$V(E)$	Gauss–Legendre, cylindrical	19.634954084936
$V(E)$	Gauss–Legendre, Cartesian	19.634954084933

Table 1: The volume of the solid. The two exact forms agree to all digits, and the four numerical values agree with them to the accuracy of each method. The Monte Carlo values use 8×10^6 samples and carry a standard error of about 4×10^{-3}.

3. The volume, as a check that the picture is right

A description of a solid is only convincing if it gives the right volume. The solid is a solid of revolution about the z-axis, so the disk method applies:

$$\begin{aligned} V(E) &= \int_0^{5/2} \pi z\,\mathrm{d}z + \int_{5/2}^{5} \pi(5-z)\,\mathrm{d}z \\ &= \pi\left[\frac{z^2}{2}\right]_0^{5/2} + \pi\left[5z - \frac{z^2}{2}\right]_{5/2}^{5} \\ &= \pi\cdot\frac{25}{8} + \pi\left(\frac{25}{2} - \frac{75}{8}\right) = \frac{25\pi}{8} + \frac{25\pi}{8} = \frac{25\pi}{4}. \end{aligned}$$

The same number comes from one cylindrical integral. At radius r the upper bowl is at $z = 5-r^2$ and the lower bowl at $z = r^2$, so the vertical thickness is

$$5 - r^2 - r^2 = 5 - 2r^2.$$

Hence

$$V(E) = \int_0^{2\pi}\int_0^{R} (5-2r^2)\,r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\left[\frac{5r^2}{2} - \frac{r^4}{2}\right]_0^{R} = 2\pi\left(\frac{25}{4} - \frac{25}{8}\right) = \frac{25\pi}{4}.$$

Both routes give

$$V(E) = \frac{25\pi}{4} \approx 19.634954.$$

Figure 2 shows the same solid with the region of integration shaded. The solid is cut by horizontal planes, and each cut is a filled disk of radius $r(z)$; the stack of disks is the solid.

4. Numerical confirmation

Table 1 lists the exact volume and four independent numerical values. The exact value comes from sympy, and the numerical values come from numpy: Monte Carlo sampling in a bounding box, Monte Carlo sampling in a cylinder, Gauss–Legendre quadrature in cylindrical coordinates, and Gauss–Legendre quadrature in Cartesian coordinates with the z integral done exactly.

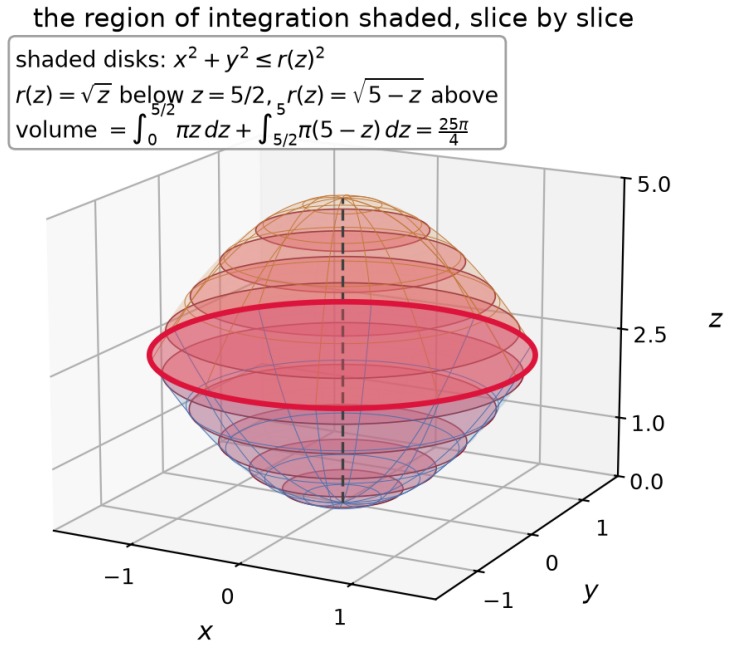


Figure 2: The region of integration shaded, slice by slice. Each red disk is the cross-section $x^2 + y^2 \le r(z)^2$ at one height z. The disks are small near the two vertices $(0, 0, 0)$ and $(0, 0, 5)$, widest at $z = 5/2$, and the crimson circle is the rim. The two paraboloids are drawn faintly so that the disks remain visible. The volume formula in the box is the integral computed in item 3.

How to solve problems of this kind

1. **Solve for the intersection first.** A solid enclosed by two surfaces is described by the curve where the surfaces meet. Set the two expressions for z equal, and solve for the horizontal variables. Here that step gives the circle $x^2 + y^2 = 5/2$.

2. **Decide which surface is on top.** Test one interior point, for example the axis. At $r = 0$ the lower bowl is at $z = 0$ and the upper bowl is at $z = 5$, so the upper bowl is the roof. This test also tells you the range of the outer variables.

3. **Look for an axis of symmetry.** When both equations use $x^2 + y^2$ only, the solid is a solid of revolution. Then cylindrical coordinates, or the disk method, replace a triple integral by a single one.

4. **Read the shape from two slices.** A horizontal slice gives the disk $x^2 + y^2 \le r(z)^2$, and a vertical slice through the axis gives two parabolas that meet at two points. Those two slices together identify the solid, and they are what a graphing device draws.

5. **Check the answer with the volume.** The volume must be positive, and it must agree with the geometry: here the solid sits inside the cylinder of radius R and height 5, whose volume is $25\pi/2 \approx 39.27$, and $25\pi/4$ is exactly half of that.

6. **Confirm with a second method.** Do the exact integral once by hand, or with sympy, and once numerically with numpy. If the two disagree, the bounds or the integrand are wrong.

Answer

E is the closed lens between the bowls $z = x^2 + y^2$ and $z = 5 - x^2 - y^2$:

$$E = \Big\{(x, y, z) : x^2 + y^2 \le \tfrac{5}{2},\ \ x^2 + y^2 \le z \le 5 - x^2 - y^2\Big\}.$$

The two bowls meet on the circle $x^2 + y^2 = 5/2$ at height $z = 5/2$, the z-axis is an axis of symmetry, and

$$V(E) = \frac{25\pi}{4} \approx 19.635.$$

A graphing device reproduces Figures 1 and 2 with the window $-2 \le x \le 2$, $-2 \le y \le 2$, $0 \le z \le 5$: the upward bowl appears as the floor of the solid and the downward bowl as its roof, and the widest circle in the picture is the rim at $z = 5/2$. The volume $25\pi/4 \approx 19.63$ is exactly half of the volume $25\pi/2$ of the cylinder $x^2 + y^2 \le 5/2$, $0 \le z \le 5$ that contains the solid, which is the quickest independent check on the answer.